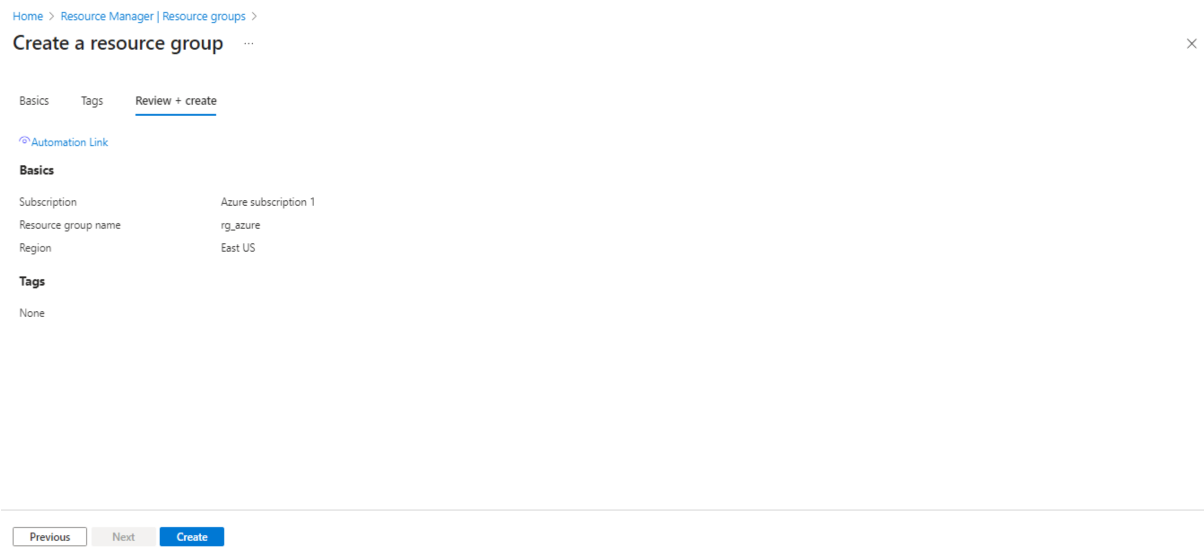
Insurance Charges Prediction

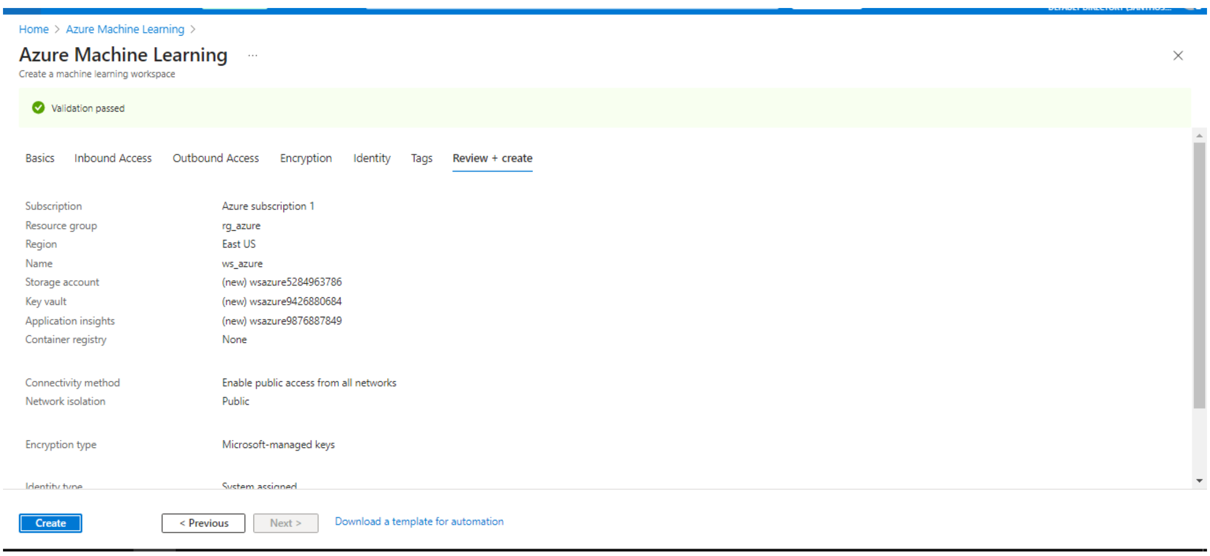
Objective:
Build and train a regression model to predict insurance charges for customers using **Azure Machine Learning**, and deploy the model for real-time predictions.

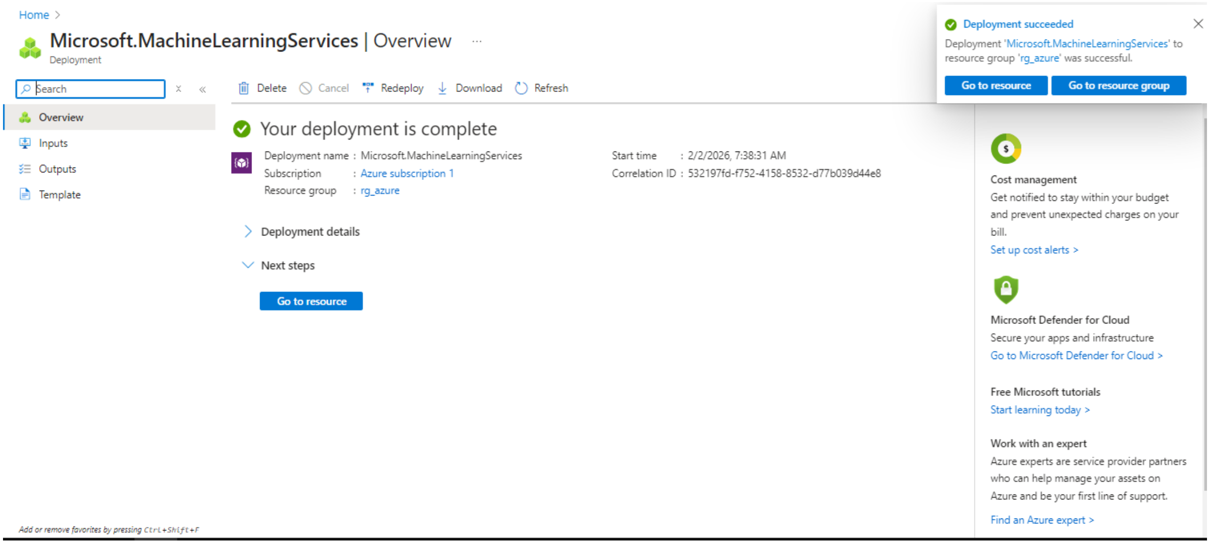
Steps performed:

Create Resource group

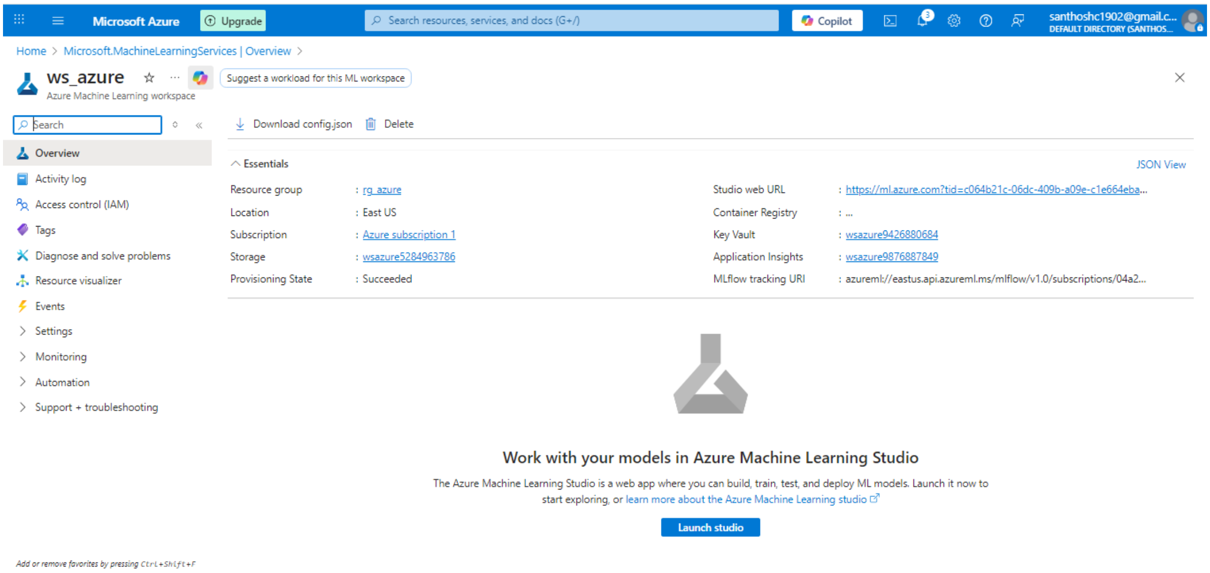


Azure ML→Create workspace

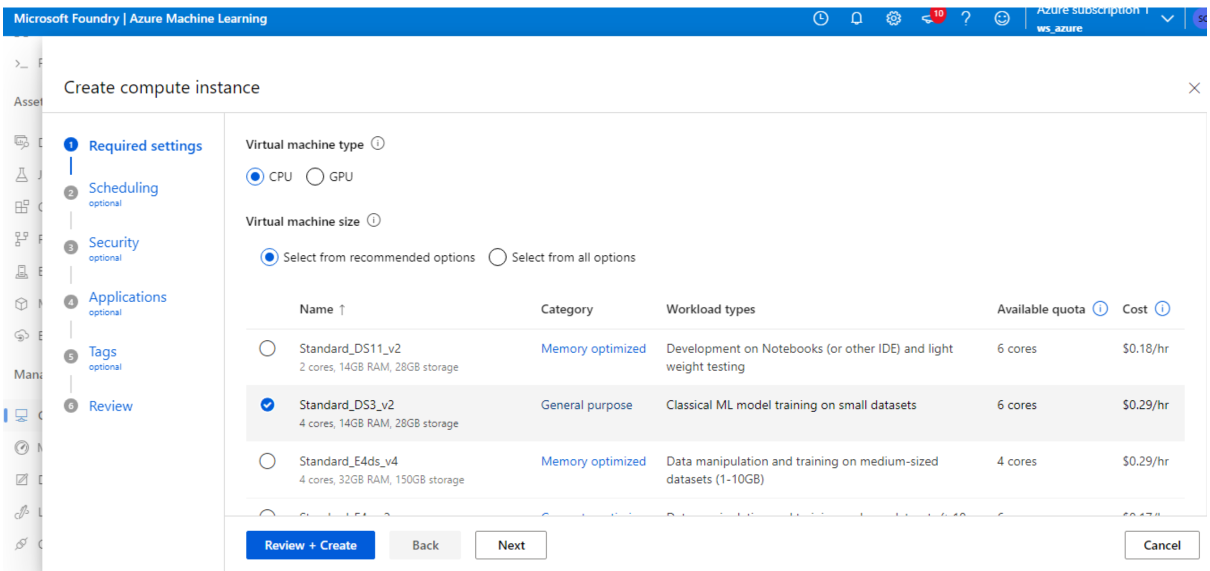


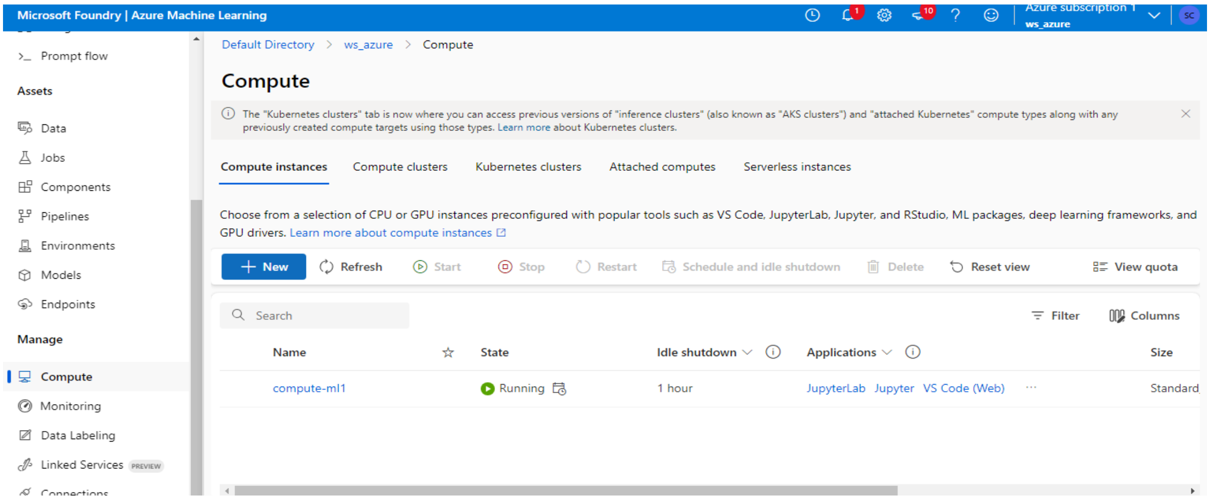


Launch ML Studio

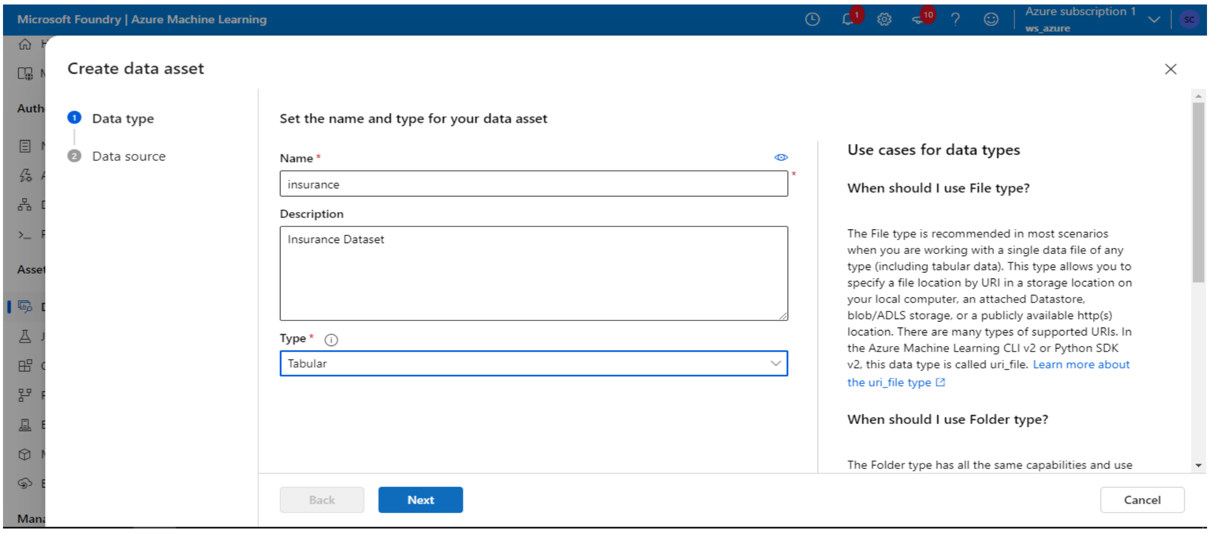


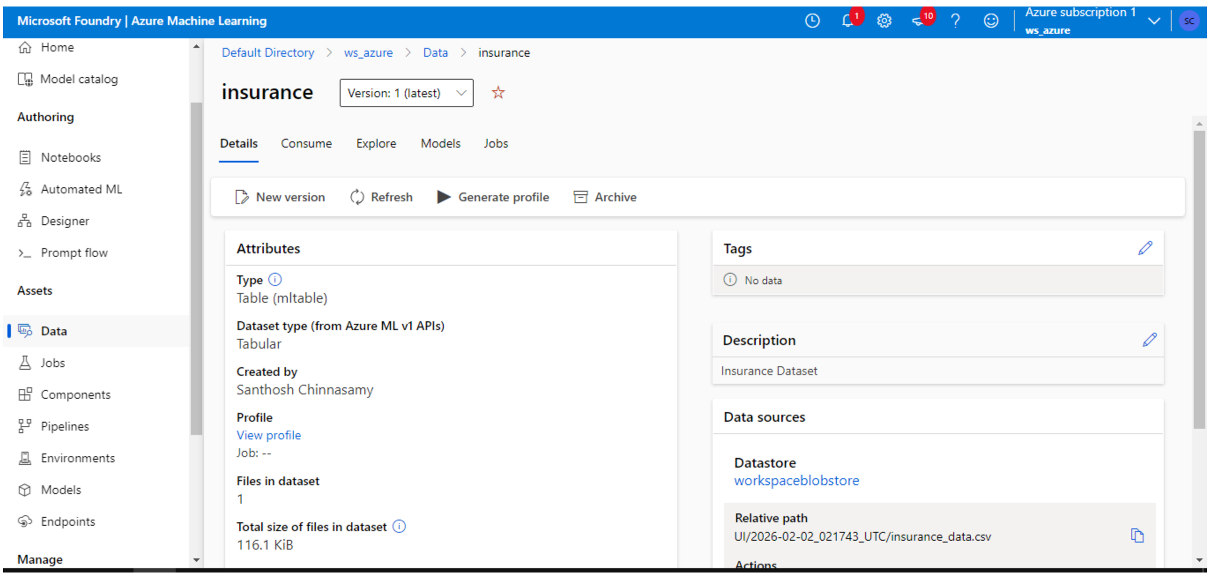
Create Compute




Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
Linked Services
Default Directory > ws_azure > Compute
Compute
The "Kubernetes clusters" tab is now where you can access previous versions of "inference clusters" (also known as "AKS clusters") and "attached Kubernetes" compute types along with any previously created compute targets using those types. Learn more about Kubernetes clusters.
Compute instances
Compute clusters
Kubernetes clusters
Attached computes
Serverless instances
Choose from a selection of CPU or GPU instances preconfigured with popular tools such as VS Code, JupyterLab, Jupyter, and RStudio, ML packages, deep learning frameworks, and GPU drivers. Learn more about compute instances
New
Refresh
Start
Stop
Restart
Schedule and idle shutdown
Delete
Reset view
View quota
Search
Filter
Columns
Name
State
Idle shutdown
Applications
Size
compute-ml1
Running
1 hour
JupyterLab
Jupyter
VS Code (Web)
Standard

Data Load


Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Create data asset
Data type
Data source
Set the name and type for your data asset
Name *
insurance
Description
Insurance Dataset
Type *
Tabular
Use cases for data types
When should I use File type?
The File type is recommended in most scenarios when you are working with a single data file of any type (including tabular data). This type allows you to specify a file location by URI in a storage location on your local computer, an attached Datastore, blob/ADLS storage, or a publicly available http(s) location. There are many types of supported URIs. In the Azure Machine Learning CLI v2 or Python SDK v2, this data type is called uri_file. Learn more about the uri_file type
When should I use Folder type?
The Folder type has all the same capabilities and use
Back
Next
Cancel


Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Home
Model catalog
Authoring
Notebooks
Automated ML
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Prompt flow
Assets
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Jobs
Components
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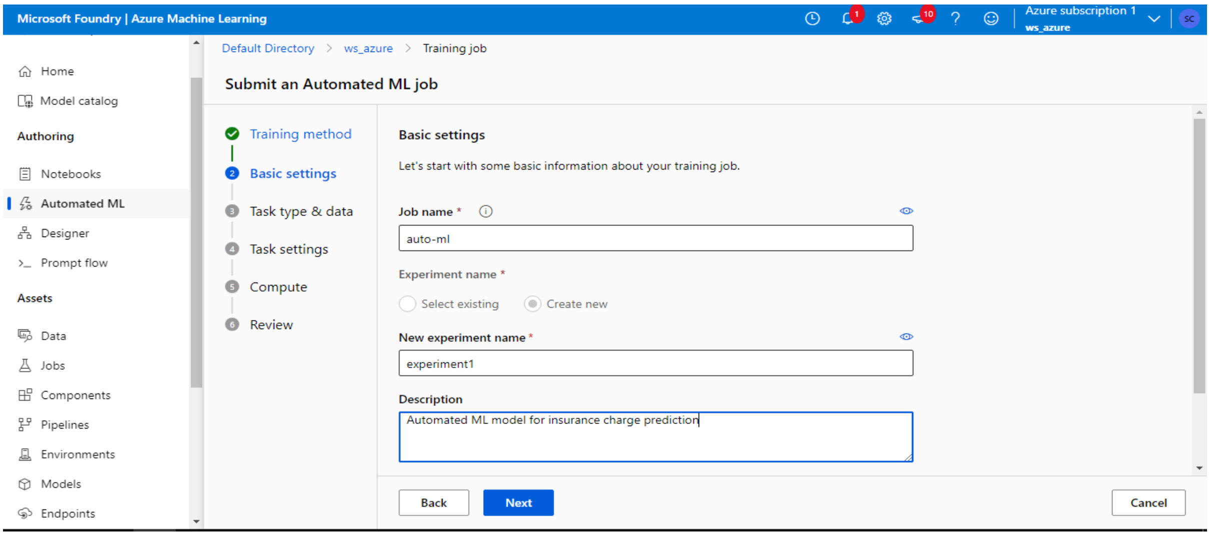
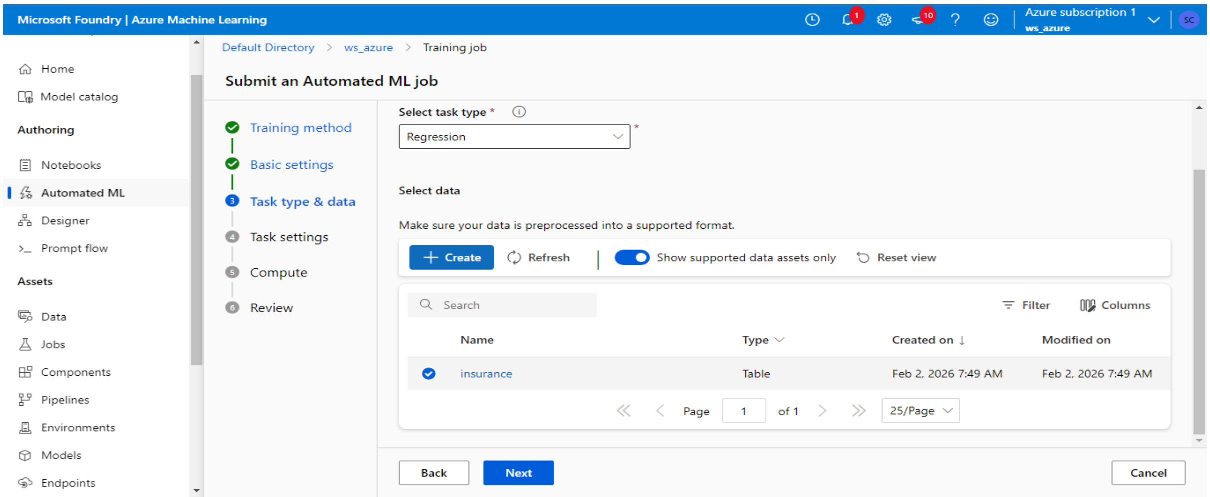
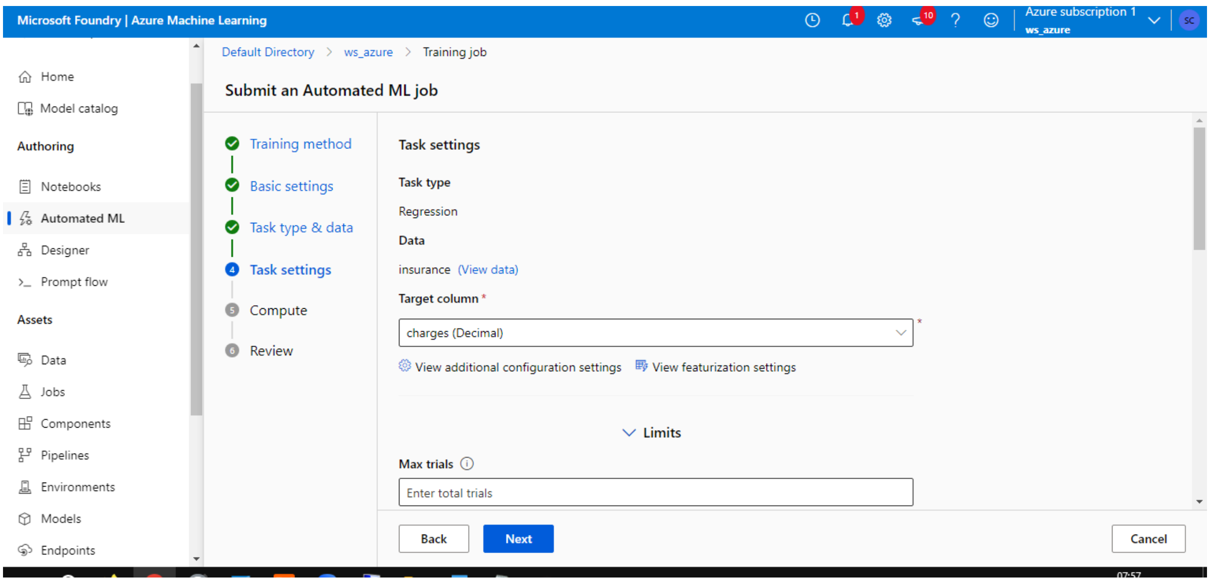
Manage
Default Directory > ws_azure > Data > insurance
insurance
Version: 1 (latest)
Details
Consume
Explore
Models
Jobs
New version
Refresh
Generate profile
Archive
Attributes
Type
Table (mltable)
Dataset type (from Azure ML v1 APIs)
Tabular
Created by
Santhosh Chinnasamy
Profile
View profile
Job: --
Files in dataset
1
Total size of files in dataset
116.1 KiB
Tags
No data
Description
Insurance Dataset
Data sources
Datastore
workspaceblobstore
Relative path
UI/2026-02-02_021743_UTC/insurance_data.csv
Actions

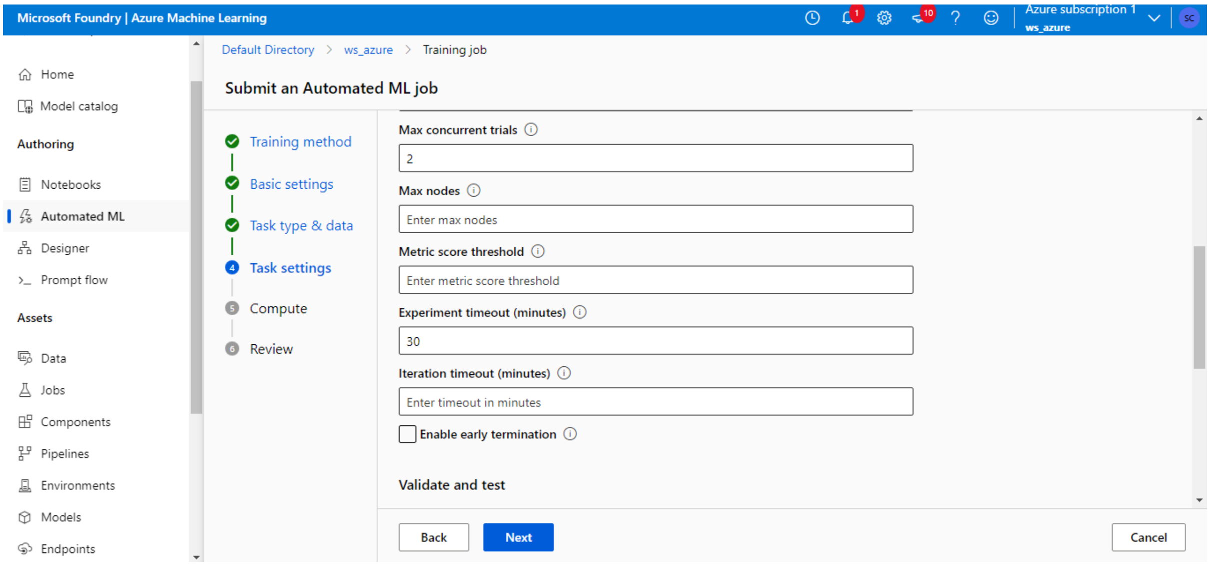
Model Training

Automated ML





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Azure subscription 1
ws_azure
Default Directory > ws_azure > Training job
Submit an Automated ML job
Home
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Automated ML
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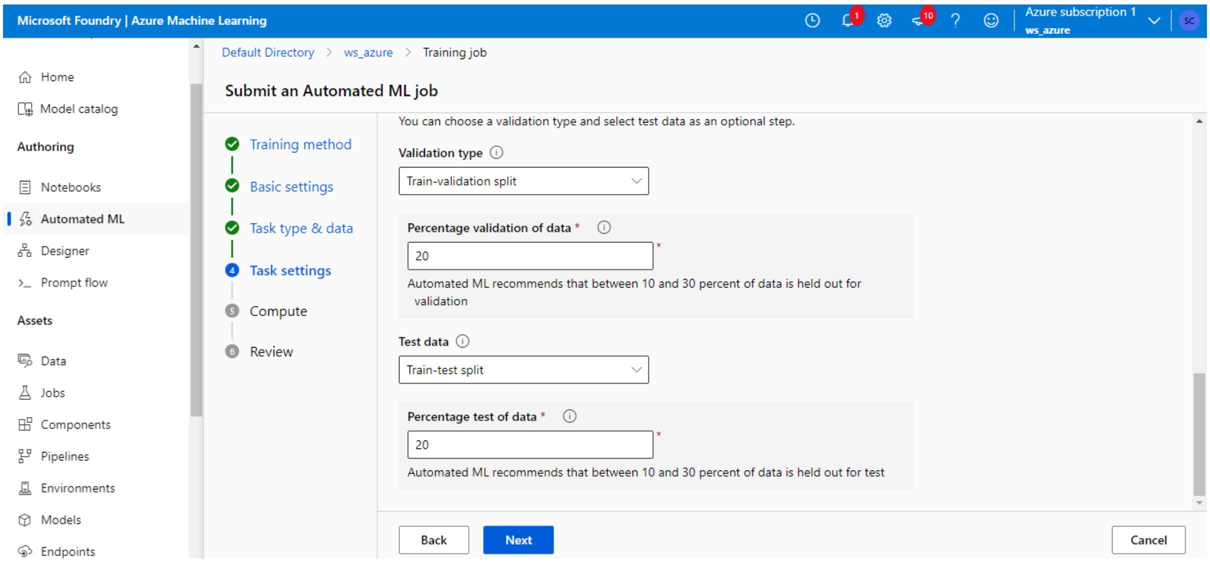
Endpoints
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Max concurrent trials
2
Max nodes
Enter max nodes
Metric score threshold
Enter metric score threshold
Experiment timeout (minutes)
30
Iteration timeout (minutes)
Enter timeout in minutes
Enable early termination
Validate and test
Back
Next
Cancel


Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Default Directory > ws_azure > Training job
Submit an Automated ML job
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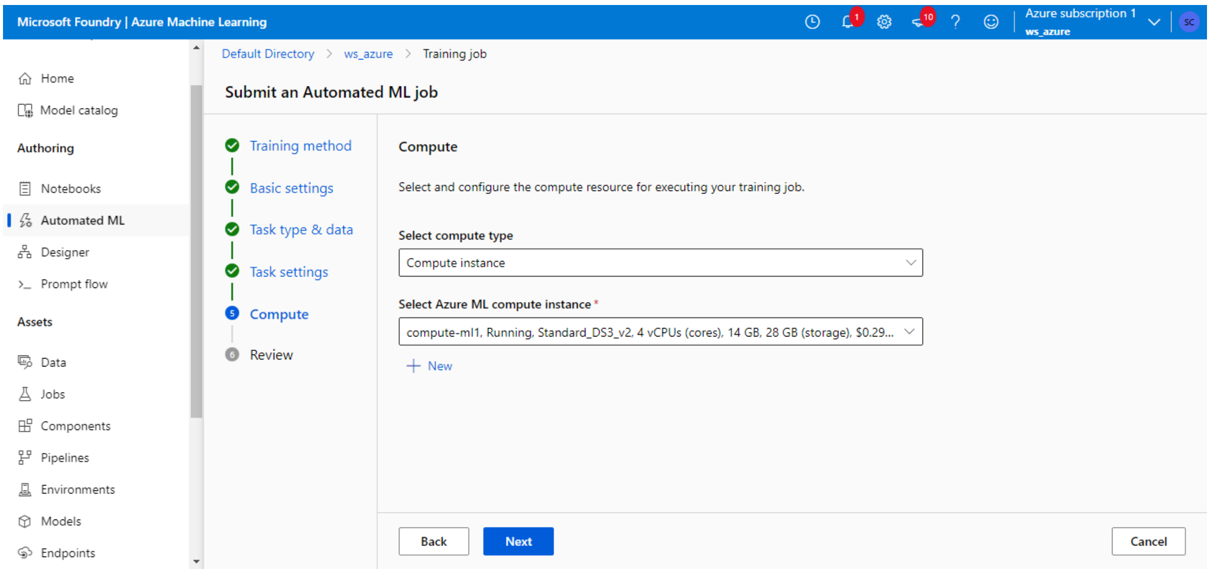
Models
Endpoints
Training method
Basic settings
Task type & data
Task settings
Compute
Review
You can choose a validation type and select test data as an optional step.
Validation type
Train-validation split
Percentage validation of data *
20
Automated ML recommends that between 10 and 30 percent of data is held out for validation
Test data
Train-test split
Percentage test of data *
20
Automated ML recommends that between 10 and 30 percent of data is held out for test
Back
Next
Cancel


Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Default Directory > ws_azure > Training job
Submit an Automated ML job
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
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Jobs
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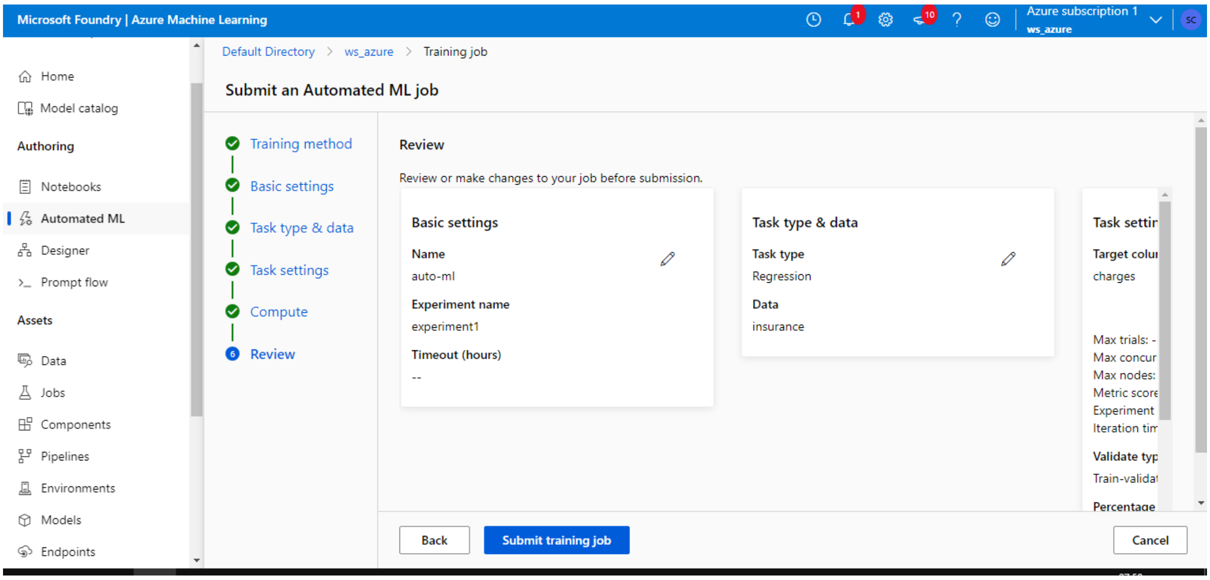
Pipelines
Environments
Models
Endpoints
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Compute
Select and configure the compute resource for executing your training job.
Select compute type
Compute instance
Select Azure ML compute instance *
compute-ml1, Running, Standard_DS3_v2, 4 vCPUs (cores), 14 GB, 28 GB (storage), $0.29...
New
Back
Next
Cancel


Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Default Directory > ws_azure > Training job
Submit an Automated ML job
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
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Pipelines
Environments
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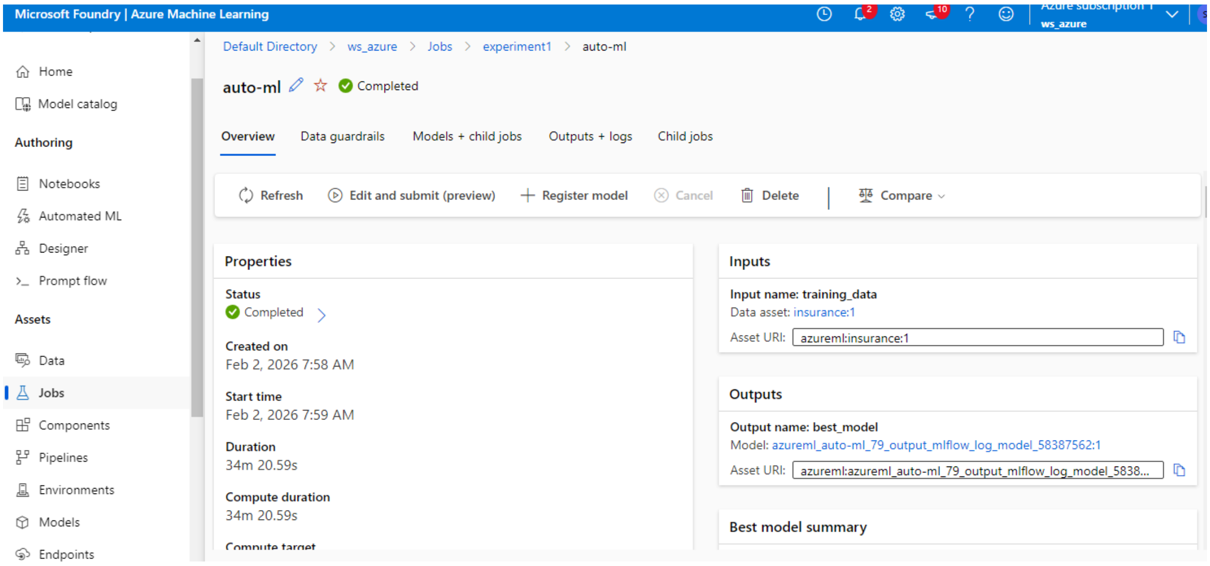
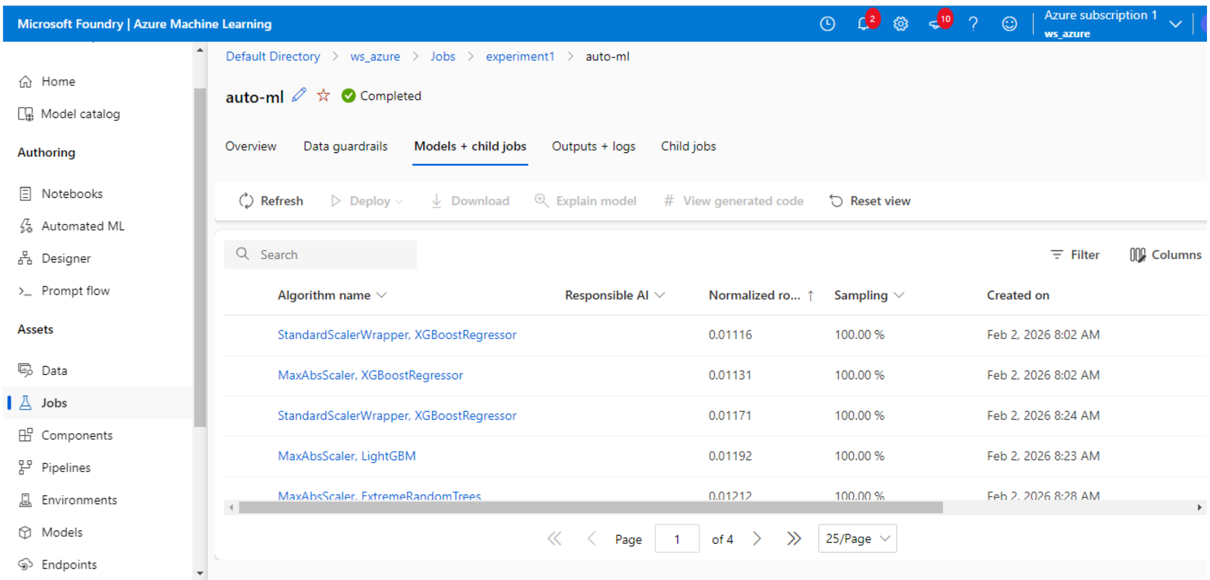
Endpoints
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Review
Review or make changes to your job before submission.
Basic settings
Name
auto-ml
Experiment name
experiment1
Timeout (hours)
--
Task type & data
Task type
Regression
Data
insurance
Task settir
Target colu
charges
Max trials: -
Max concur
Max nodes:
Metric score
Experiment
Iteration tim
Validate typ
Train-validat
Percentage
Back
Submit training job
Cancel


Microsoft Foundry | Azure Machine Learning
ws_azure
Default Directory > ws_azure > Jobs > experiment1 > auto-ml
auto-ml
Completed
Overview
Data guardrails
Models + child jobs
Outputs + logs
Child jobs
Refresh
Edit and submit (preview)
Register model
Cancel
Delete
Compare
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Properties
Status
Completed
Created on
Feb 2, 2026 7:58 AM
Start time
Feb 2, 2026 7:59 AM
Duration
34m 20.59s
Compute duration
34m 20.59s
Inputs
Input name: training_data
Data asset: insurance:1
Asset URI: azureml:insurance:1
Outputs
Output name: best_model
Model: azureml_auto-ml_79_output_mlflow_log_model_58387562:1
Asset URI: azureml:azureml_auto-ml_79_output_mlflow_log_model_5838...
Best model summary


Microsoft Foundry | Azure Machine Learning
Azure subscription 1
ws_azure
Default Directory > ws_azure > Jobs > experiment1 > auto-ml
auto-ml
Completed
Overview
Data guardrails
Models + child jobs
Outputs + logs
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Download
Explain model
View generated code
Reset view
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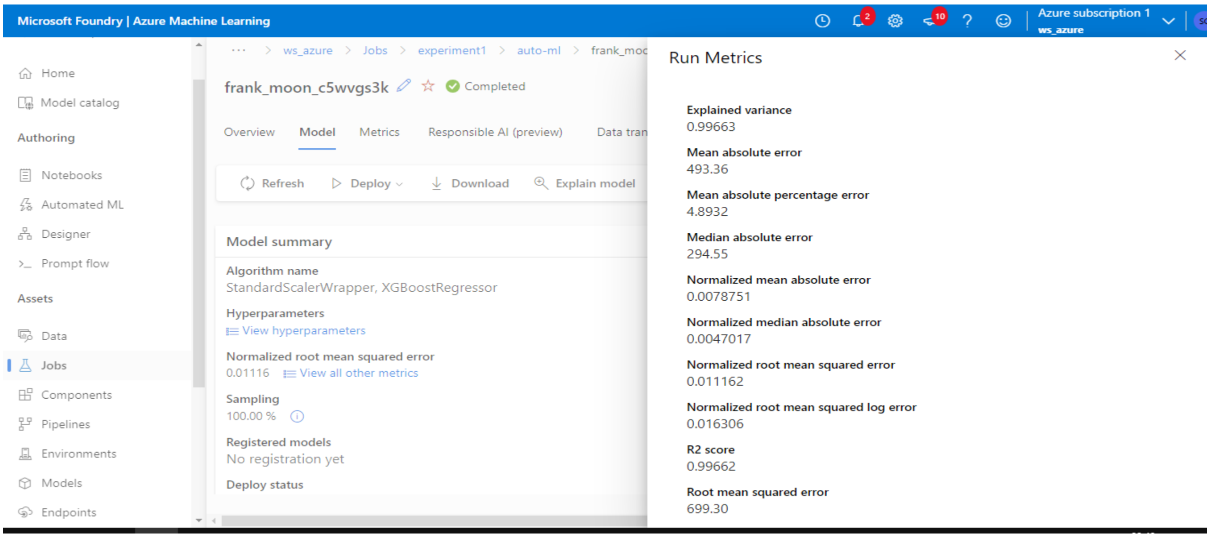
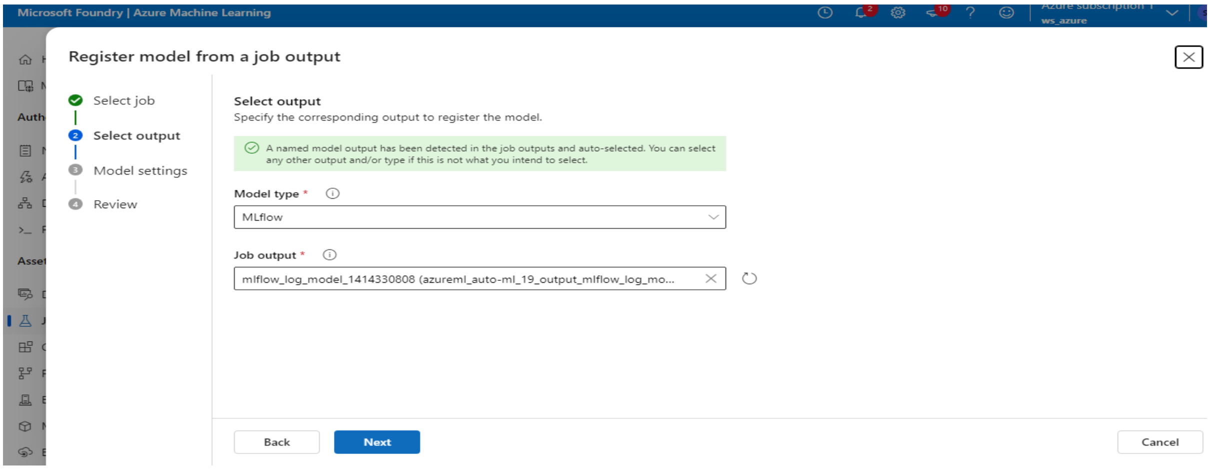
Components
Pipelines
Environments
Models
Endpoints
Search
Filter
Columns
Algorithm name
Responsible AI
Normalized ro...
Sampling
Created on
StandardScalerWrapper, XGBoostRegressor 0.01116 100.00 % Feb 2, 2026 8:02 AM
MaxAbsScaler, XGBoostRegressor 0.01131 100.00 % Feb 2, 2026 8:02 AM
StandardScalerWrapper, XGBoostRegressor 0.01171 100.00 % Feb 2, 2026 8:24 AM
MaxAbsScaler, LightGBM 0.01192 100.00 % Feb 2, 2026 8:23 AM
Page 1 of 4
25/Page

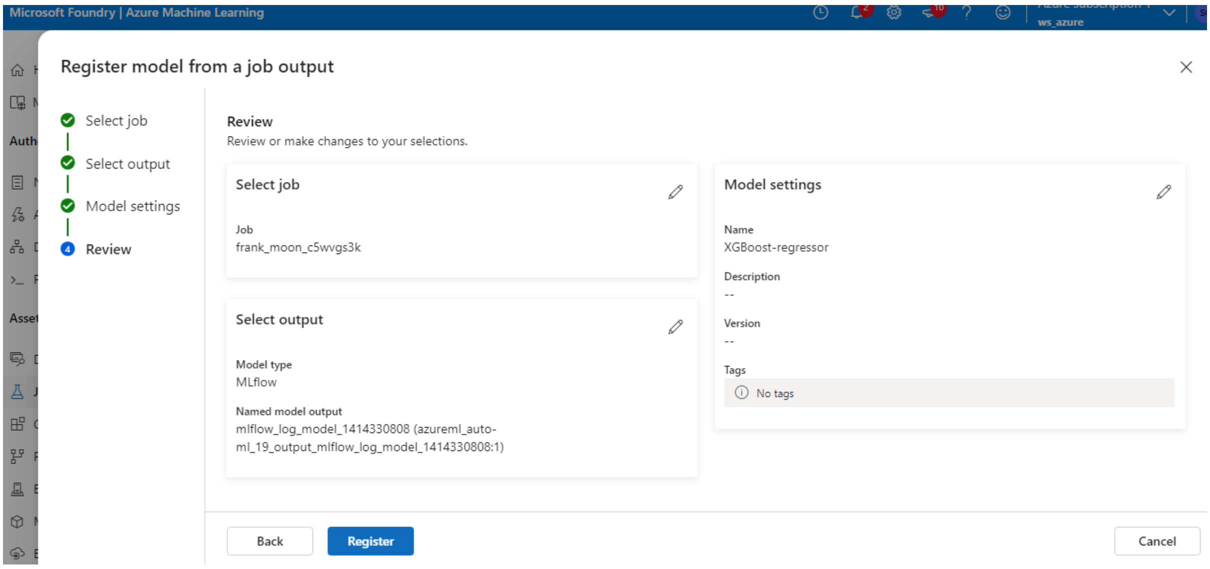
Model Selection

XGBoost Regressor has been selected and registered

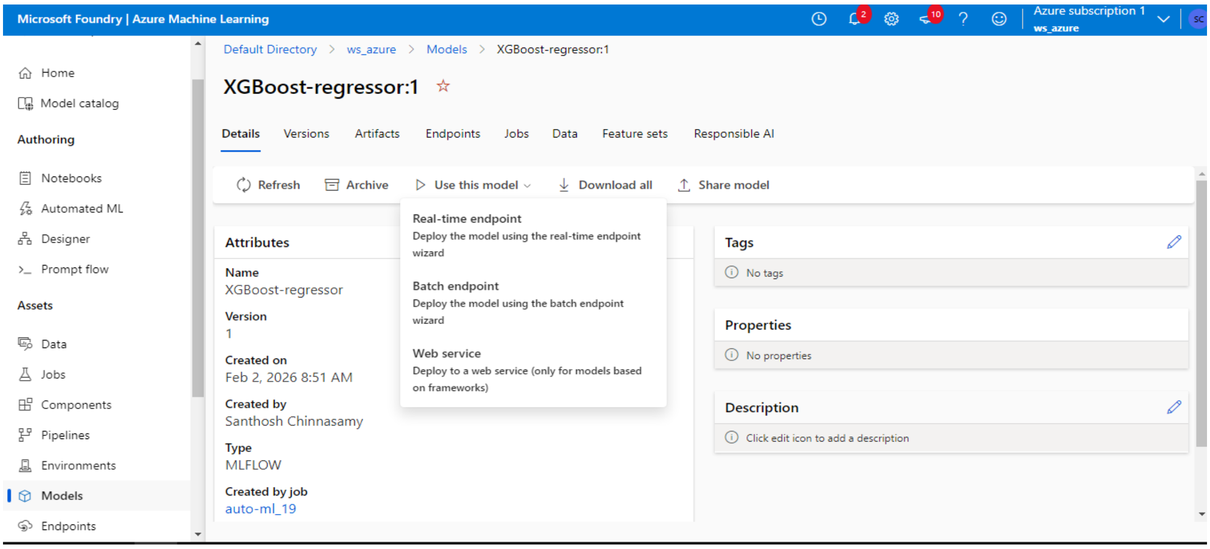


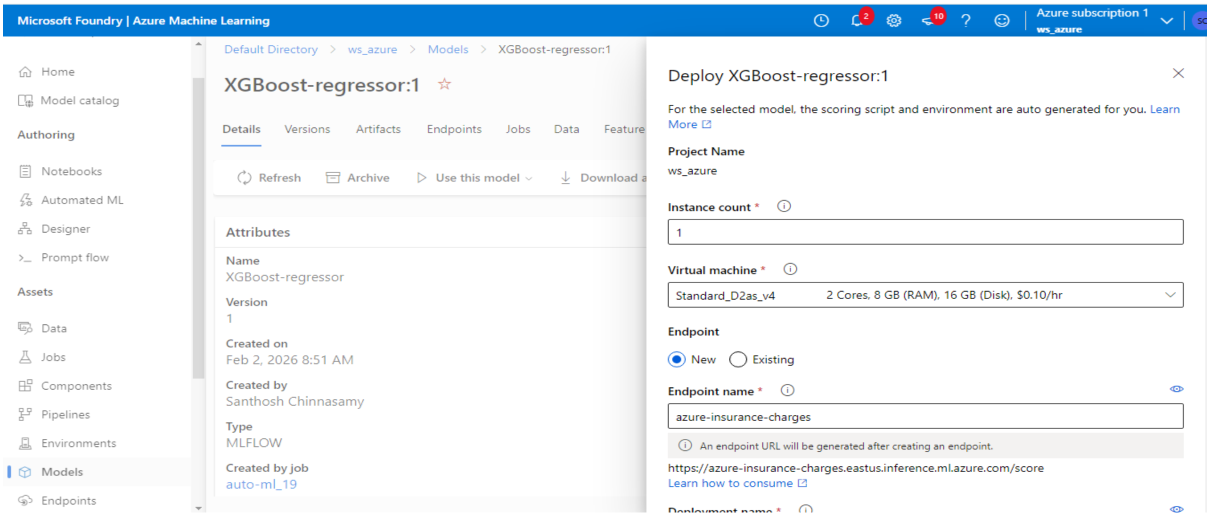
Register model

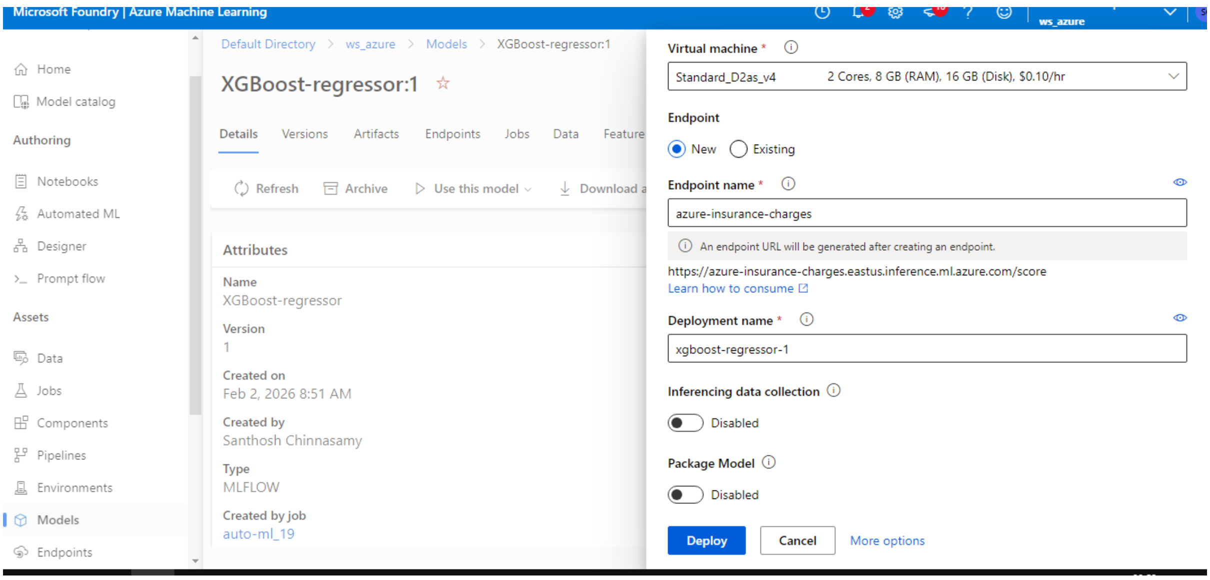


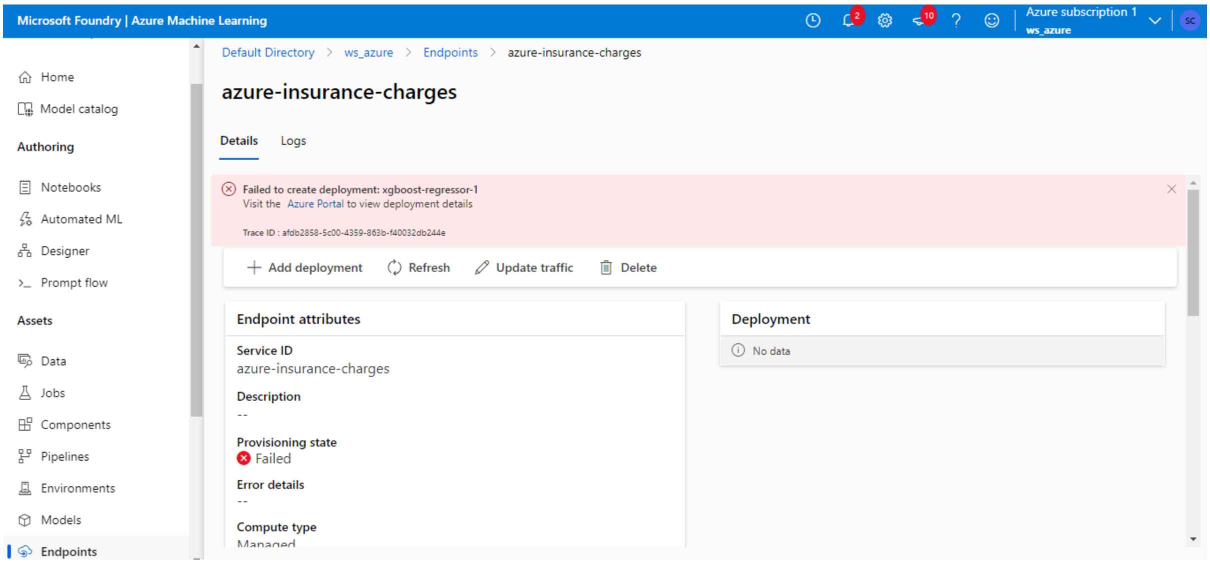


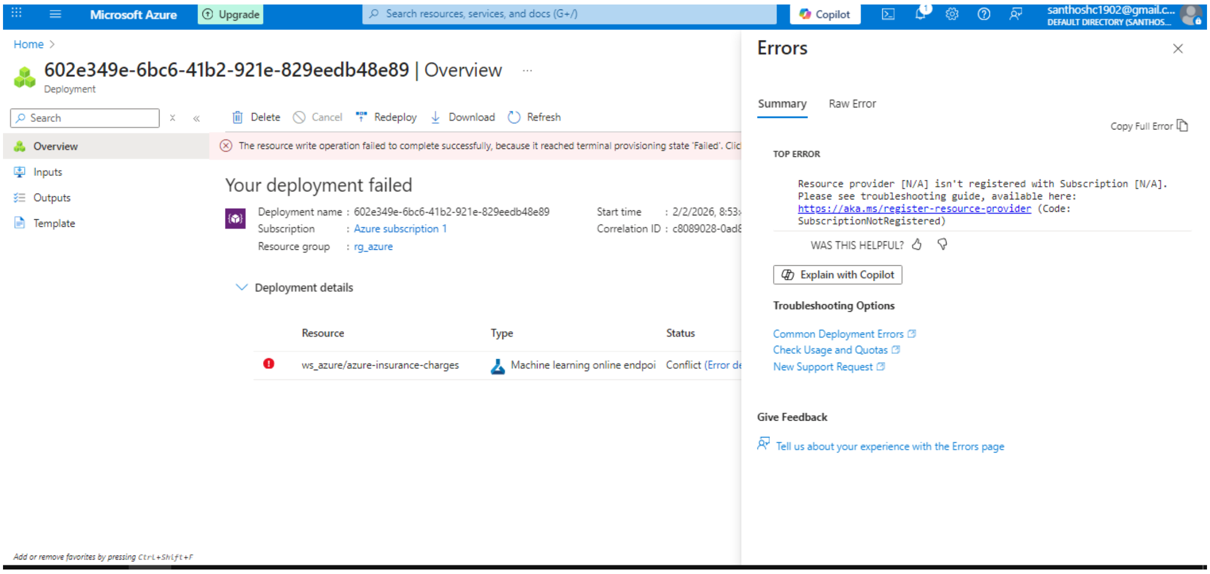
Model Deployment









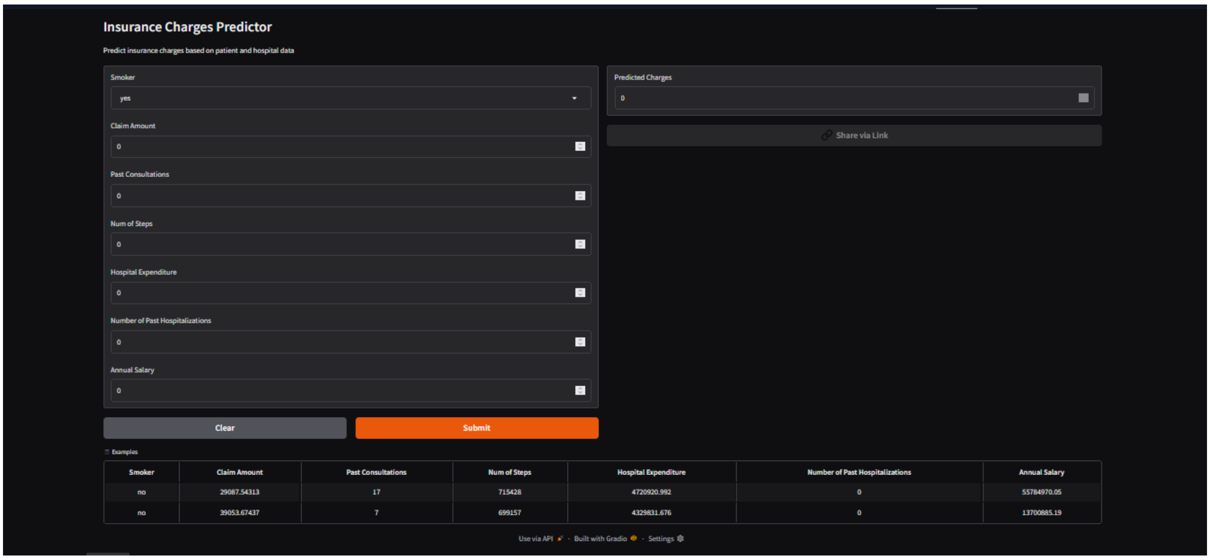


Cloud Deployment Workaround

Due to limitations in the Azure ML free trial, deployment of the model and generation of the endpoint/API key was not possible. As a workaround:

- The regression model was trained in Jupyter Notebook and saved as a .pkl file
- The saved model was then deployed on Hugging Face Spaces using Gradio for an interactive web interface

Live demo: Insurance Charges Prediction


Insurance Charges Predictor
Predict insurance charges based on patient and hospital data
Smoker
yes
Claim Amount
0
Past Consultations
0
Num of Steps
0
Hospital Expenditure
0
Number of Past Hospitalizations
0
Annual Salary
0
Predicted Charges
0
Share via Link
Clear
Submit
Examples
Smoker | Claim Amount | Past Consultations | Num of Steps | Hospital Expenditure | Number of Past Hospitalizations | Annual Salary
no | 29087.54313 | 17 | 715428 | 4720920.992 | 0 | 55784970.05
no | 39053.67437 | 7 | 699157 | 4329831.676 | 0 | 13700885.19
Use via API · Built with Gradio · Settings